

### ■ 1.7.3 Vectors and Matrices

Vectors and matrices in *Mathematica* are simply represented by lists and lists of lists, respectively.

$$\{a, b, c\} \quad \text{vector } (a, b, c)$$

$$\{\{a, b\}, \{c, d\}\} \quad \text{matrix } \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The representation of vectors and matrices by lists.

This is a  $2 \times 2$  matrix.

```
In[1]:= m = {{a, b}, {c, d}}
Out[1]= {{a, b}, {c, d}}
```

Here is the first row.

```
In[2]:= m[[1]]
Out[2]= {a, b}
```

Here is the element  $m_{12}$ .

```
In[3]:= m[[1,2]]
Out[3]= b
```

This is a two-component vector.

```
In[4]:= v = {x, y}
Out[4]= {x, y}
```

$p$  and  $q$  are treated as scalars.

```
In[5]:= p v + q
Out[5]= {q + p x, q + p y}
```

Vectors are added component by component.

```
In[6]:= v + {xp, yp} + {xpp, ypp}
Out[6]= {x + xp + xpp, y + yp + ypp}
```

This takes the dot (“scalar”) product of two vectors.

```
In[7]:= {x, y} . {xp, yp}
Out[7]= x xp + y yp
```

You can also multiply a matrix by a vector.

```
In[8]:= m . v
Out[8]= {a x + b y, c x + d y}
```

Or a matrix by a matrix.

```
In[9]:= m . m
Out[9]= {{a2 + b c, a b + b d}, {a c + c d, b c + d2}}
```

Or a vector by a matrix.

```
In[10]:= v . m
Out[10]= {a x + c y, b x + d y}
```

This combination makes a scalar.

```
In[11]:= v . m . v
Out[11]= x (a x + c y) + y (b x + d y)
```

The way *Mathematica* uses lists to represent vectors and matrices, you never have to distinguish between “row” and “column” vectors.

|                                       |                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>Table[f, {i, n}]</code>         | build a length- $n$ vector by evaluating $f$ with $i=1, i=2, \dots, i=n$                                    |
| <code>Array[a, n]</code>              | build a length- $n$ vector of the form $\{a[1], a[2], \dots\}$                                              |
| <code>list[[i]]</code>                | give the $i^{\text{th}}$ element in the vector $list$                                                       |
| <code>ColumnForm[list]</code>         | display the elements of $list$ in a column                                                                  |
| <code>Table[f, {i, m}, {j, n}]</code> | build an $m \times n$ matrix by evaluating $f$ with $i$ ranging from 1 to $m$ and $j$ ranging from 1 to $n$ |
| <code>Array[a, {m, n}]</code>         | build an $m \times n$ matrix with $i, j^{\text{th}}$ element $a[i, j]$                                      |
| <code>IdentityMatrix[n]</code>        | generate an $n \times n$ identity matrix                                                                    |
| <code>DiagonalMatrix[list]</code>     | generate a square matrix with the elements in $list$ on the diagonal                                        |
| <code>list[[i]]</code>                | give the $i^{\text{th}}$ row in the matrix $list$                                                           |
| <code>list[[i, j]]</code>             | give the $i, j^{\text{th}}$ element in matrix $list$                                                        |
| <code>MatrixForm[list]</code>         | display $list$ in matrix form                                                                               |

#### Functions for vectors and matrices.

This builds a  $3 \times 3$  matrix  $s$  with elements  $s_{ij} = i + j$ .

```
In[12]:= s = Table[i+j, {i, 3}, {j, 3}]
Out[12]= {{2, 3, 4}, {3, 4, 5}, {4, 5, 6}}
```

This displays  $s$  in standard two-dimensional matrix format.

```
In[13]:= MatrixForm[s]
      2  3  4
      3  4  5
Out[13]//MatrixForm= 4  5  6
```

This gives a vector with “symbolic elements”. You can use this in deriving general formulae that are valid with any choice of vector components.

```
In[14]:= Array[a, 4]
Out[14]= {a[1], a[2], a[3], a[4]}
```

This gives a  $3 \times 2$  matrix with symbolic elements. Chapter 2.1.14 will discuss how you can produce other kinds of elements with Array.

```
In[15]:= Array[p, {3, 2}]
Out[15]= {{p[1, 1], p[1, 2]}, {p[2, 1], p[2, 2]},
          {p[3, 1], p[3, 2]}}
```

This generates a  $3 \times 3$  diagonal matrix.

```
In[16]:= DiagonalMatrix[{a, b, c}]
Out[16]= {{a, 0, 0}, {0, b, 0}, {0, 0, c}}
```

|                                       |                                        |
|---------------------------------------|----------------------------------------|
| $c \ m$                               | multiply by a scalar                   |
| $a \ . \ b$                           | matrix product                         |
| Inverse[m]                            | matrix inverse                         |
| Det[m]                                | determinant                            |
| Transpose[m]                          | transpose                              |
| Eigenvalues[m]                        | eigenvalues                            |
| Eigenvectors[m]                       | eigenvectors                           |
| Eigenvalues[N[m]], Eigenvectors[N[m]] | numerical eigenvalues and eigenvectors |

Some mathematical operations on matrices.

Here is a  $2 \times 2$  matrix of symbolic variables, that was defined above.

```
In[17]:= m
Out[17]= {{a, b}, {c, d}}
```

This gives its determinant.

```
In[18]:= Det[m]
Out[18]= -(b c) + a d
```

Here is the transpose of m.

```
In[19]:= Transpose[m]
Out[19]= {{a, c}, {b, d}}
```

This gives the inverse of **m** in symbolic form.

```
In[20]:= Inverse[m]
Out[20]= {{(d)/(- (b c) + a d), -(b)/(- (b c) + a d)},
           {-(c)/(- (b c) + a d), (a)/(- (b c) + a d)}}
```

Here is a particular  $3 \times 3$  rational matrix known as a “Hilbert matrix”.

```
In[21]:= h = Table[1/(i+j+1), {i, 3}, {j, 3}]
Out[21]= {{1/3, 1/4, 1/5}, {1/4, 1/5, 1/6}, {1/5, 1/6, 1/7}}
```

This gives its inverse of **h**.

```
In[22]:= Inverse[h]
Out[22]= {{300, -900, 630}, {-900, 2880, -2100},
           {630, -2100, 1575}}
```

Taking the dot product of the inverse with the original matrix gives the identity matrix.

```
In[23]:= % . h
Out[23]= {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}}
```

Here is a  $3 \times 3$  matrix.

```
In[24]:= r = Table[i+j+1, {i, 3}, {j, 3}]
Out[24]= {{3, 4, 5}, {4, 5, 6}, {5, 6, 7}}
```

**Eigenvalues** gives the eigenvalues of the matrix.

```
In[25]:= Eigenvalues[r]
Out[25]= {0, (15 + Sqrt[249])/2, (15 - Sqrt[249])/2}
```

This gives a numerical approximation to the matrix.

```
In[26]:= rn = N[r]
Out[26]= {{3., 4., 5.}, {4., 5., 6.}, {5., 6., 7.}}
```

Here are numerical approximations to the eigenvalues.

```
In[27]:= Eigenvalues[rn]
Out[27]= {15.3899, -0.389867, 4.6843 10-18}
```

Section 3.7 discusses a number of other matrix operations that are built in to *Mathematica*.